

Chapter 9 Build a Solution: Design the Run-time Interface

The Run-time Interface is for intuitive control and monitoring of the solution in the running status. With the Run-time Interface, the end users, i.e., the operators at the manufacturing site, can view the inspection results in real time and determine whether to stop running the solution—even the corresponding assembly line at large—in particular cases. This section introduces the features of the Run-time Interface Design window and how to customize the interface to suit specific requirements.

9.1 Overview of the Run-time Interface Design Window

The Run-time Interface Design window enables the design and customization of the Run-time Interface. This section provides an overview of the graphic user interface (GUI) of the window and the core function of each area.

Click the  in the toolbar on the main page to go to the Run-time Interface Design window.

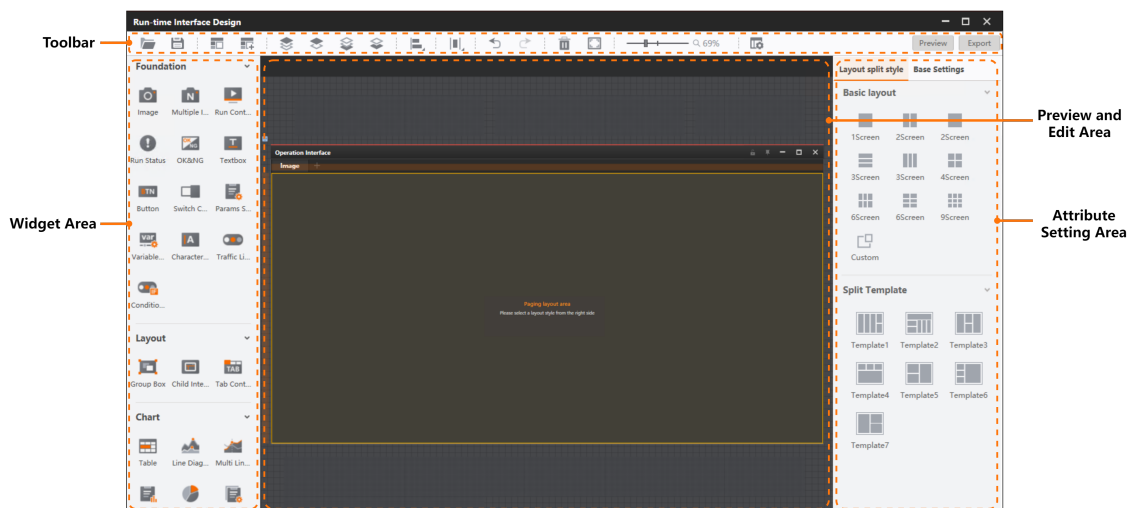


Figure 9-1 Window Overview

The following table describes the functions of each area.

Table 9-1 Run-time Interface Design Introduction

Function Area	Description
Toolbar	Provides functions such as running page saving, layer setting, deletion, and export. For details, see <i>Toolbar Introduction</i> .
Widget Area	Contains widgets available to use on the running page. For details, see <i>Widget Configuration Reference</i> .
Preview and Edit Area	Displays the currently-designed running page. Allows you to create widgets and edit the added widgets. For details, see <i>Basic Operations on Widgets</i> .
Attribute Setting Area	Allows you to set attributes (such as layout, data source, color, and font) of the widget selected in the preview and edit area.

Toolbar Introduction

You can use the following functions in the toolbar.

-  : open a saved Run-time Interface design from a local directory.
-  : save the current design. For first-time saving, you save a design file (.vmd).
-  : open the design template window. You can apply a design template and delete saved templates.
-  : save the current Run-time Interface as a template. You need to enter the **Name**, **Created By**, **Usage Scenarios**, and **Description**, then click **Confirm**.
-     : move the selected widgets' layer. They mean bring to front, bring forward, send to back, and send backward respectively.
-  : align selected widgets. The alignment area is based on the minimal bounding box area that contains all selected widgets. You can align them to the left, center, right, top, vertically center, top, or bottom.

Note

Select at least 2 widgets.

-
-  : justify the widgets vertically or horizontally. The justify area is based on the minimal bounding box area that contains all selected widgets.
-

Note

Select at least 3 widgets; otherwise, it will not justify the distribution.

-
-  : undo one recent operation.
 -  : redo one recent operation.
 -  : remove the select widgets.
 -  : zoom the editing area to adapt to the window size.
-

Note

When you open the design window, this function is called for once by default.

-  84% : click and drag the slider to zoom in or out the display ratio.
-  : set whether to add a fixed column at the top, bottom, left or right side of the Run-time Interface.
- **Preview**: click to preview the Run-time Interface.
- **Export**: click to compile and export the solution into a project package file. For details, see *[Run a Solution by Exporting the Operation Window](#)*.

Note

- You can right-click a widget to delete, undo, redo, or move its layer.
 - You can copy and paste selected widgets through keyboard shortcuts Ctrl+C (for copying) and Ctrl+V (for pasting).
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9.2 Design the Run-time Interface

You can design the Run-time Interface based on the business requirements to help the end users monitor and control the solution execution easily and more intuitively, thus improving their work efficiency.

9.2.1 General Design Approach

By default, the Run-time Interface is empty, and you need to customize it in the **Run-time Interface Design** window as needed.

The overall design approach is:

1. Use a template or not: in the **Run-time Interface Design** window, click  to check if the template satisfies your requirements. If yes, you can select it and modify the design as needed; if not, proceed to the next step.
2. Set fixed column layout: click  to set the required fixed column at the top/bottom/left/right side of the Run-time Interface.
3. Set the layout: select the canvas and set the overall layout in the attribute setting area.
4. Add and configure widgets: click and drag required widgets into the fixed columns or the canvas, and configure the input source and other settings.

Note

When setting a widget, you can set its position, and alignment with controls in the toolbar.

9.2.2 Basic Operations on Widgets

The basic functional unit in an operation window is the widget. To design a well-functioning window, add multiple widgets and configure their parameters.

In the Widget Area, click and drag a required widget, move it into the Preview and Edit Area, and drop. Once added, you modify its position and size as needed and configure the Data Source and other settings in the Attribute Setting Area.

See the following video for the operation example.

9.2.3 Design Example

This section uses a real solution as an example to explain the whole process of designing a Run-time Interface.

Before You Start

Ensure you have built a solution before proceeding.



Note

This section uses the solution below as an example to explain the building process.

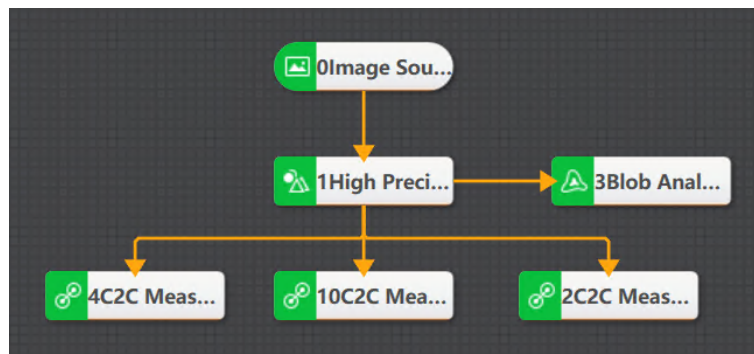


Figure 9-2 Example Solution

Steps

1. Click  to open the **Run-time Interface Design** window.
2. Click , and check **TopFixed Bar** and **BottomFixed Bar**, then click **Confirm**.
3. Design the TopFixed Bar.
 - 1) Select the top bar and set **Split Template**. It is Template1 by default. Here, the design employs the default template.
 - 2) Drag and drop the **Textbox** widget into the left column to the topfixed bar, and configure related settings.

The following only explains the changed parameters. Other parameters remain the default settings.

- **Text:** enter "Castings C2C Dist."
 - **Auto Fill:** enable it.
 - **Font:** set the font size to "48" and the font style to "bold".
- 3) Drag and drop the **Run Control** widget into the middle column of the bar, and configure related settings.

 Note

Use the default settings for this widget.

- 4) Drag and drop the **OK&NG** widget into the middle column of the bar, place it on the right side of the **Run Control** widget, and configure its settings.

 Note

Set **Data Source** to subscribe **Flow1** → **C2C Measurement2** → **ModuStatus** . Use the default settings for other parameters.

- 5) Drag and drop the **Button** widget into the middle column of the bar, place it on the right side of the **OK&NG**, and configure its setting.

Set the following parameters and use the default value for others.

- **Data Source:** click **Config** to set it, set the **Type** to Solution, set **Method** to Save, then click **Confirm**.
- **Show Text:** set to "Save Sol."

Refer to the following video for the sub-steps above.

4. Design the Bottomfixed Bar. Select the bottom bar, and in the **Select List** displayed on the right side, select the devices you want to view, for example, the global camera, TCP client, VB2230, etc.
5. Set the layout of the canvas.
- 1) Select the canvas and select "Template5" in the **Split Template**.
 - 2) Modify the grid with the triangle controls on the border.
 - 3) Select the area on the upper left and select "2Screen" in the **Basic Layout**.
 - 4) Select the area on the bottom left and select "2Screen" in the **Basic Layout**.
 - 5) Set the screens from the previous steps into 2 "2Screens".
 - 6) Select the middle 2 screens and set them to "1Screen" to merge them.
 - 7) Modify the area size with the triangle controls on the border to make the 3 areas on the bottom left the same size.

Refer to the following video for the substeps above.

6. Select the left of the canvas for image display. Drag and drop the **Image** widget into the 5 screens created from the previous step, and configure related settings.

The following only explains the changed parameters. Other parameters remain the default settings.

- **Data Source:** set them to **Flow1** → **Image Source1** → **OutputImage** .
- **Graphical Configuration:** for the screen on the top left, check all graphics in C2C Measurement1, C2C Measurement2, and C2C Measurement3. For the screen on the top middle, check all graphics in Blob Analysis1. For the screen on the bottom left,

check all graphics in C2C Measurement1, C2C Measurement2, and C2C Measurement3 respectively.

- **Show ToolsBar:** set to disable for all.

Refer to the following video for the substeps above.

7. Set the right side of the canvas for viewing live status and data. Drag and drop the **Table** widget onto the right side and configure related settings.

The following only explains the changed parameters. Other parameters remain the default settings.

- **Rows:** set it to 22.
- **Columns:** set it to 5.
- **Line Title:** set it "Result_Display".
- **Data Source:** click **Config** to set the data source for each row.
 - **Column:** set to CenterX, CenterY, Runtime, Angle, and Dist respectively.
 - **Data Type:** set to float.
 - **Data Source:** set to parameters in C2C Measurement1, CircleAnnulus1CenterX, CircleAnnulus1CenterY, Time(ms), Angle, and Distance.

Refer to the following video for the substeps above.

What to do next

Open the Run-time Interface, run the solution for once or continuously, and the running status is as follows.



Figure 9-3 Operation Window

9.3 Guide to Runtime Interface Widgets

An Runtime Interface consists of canvas(es), fixed bar(s), and widget(s). The widgets are for interacting with the client, while the fixed bars and canvas are for organizing widgets and (or) displaying device status.

9.3.1 Canvas and Fixed Bars

When designing the Runtime Interface, you can configure the canvas and fixed bars with the Preview and Edit area and Attribute Setting area.

Canvas

You can add multiple canvases at the tab bar above the canvas.

- Add a canvas: in the canvas tab bar, click  to create a new canvas.
- Change a canvas' title: click a tab and enter the desired title into **Content** in the Base Settings on the right.
- Delete a canvas: right-click a tab and select **Delete**.

The canvas is in 1screen layout by default. Click the canvas and set the layout or apply a split template.

- Layout split style: configure settings in basic layout and split template.

Note

- When you set the layout, you can click and drag the triangle to adjust the size of each area.
 - After you set the layout split, you can select one or multiple areas and split them again.
-

Basic layout

Select the basic layout of the canvas. You can click **Custom** to set the rows and columns and keep the number of each between 1 and 10.

Split Template

Apply a split template to the canvas.

- Base Settings

Show Grid

When there are multiple screens on the canvas, you can enable this feature to display the grid line.

Grid Line Color

Background

Set the color of the grid line and the whole background.

TopFixed/LeftFixed/RightFixed Bar

The fixed bar on top, left, and right is similar to the canvas. You can drag and drop different widgets into these areas.

Their configurations are similar to that of the canvas. For details, see [Canvas](#).



Note

In the **Base Settings**, you can set the width of the leftfixed and rightfixed bars.

BottomFixed Bar

You can view the connection status of cameras, communication devices, controllers, and other devices. At most, you can view 20 devices at the same time.

Select List: select devices whose connection status you want to view. The available devices in the select list are those you added in **Manage Cameras in Camera Set** and **Communication Management**.

9.3.2 Widget Configuration Reference

The Runtime Interface consists of multiple widgets, which are different in functions and attribute settings.

The following table displays the widgets in the Runtime Interface in detail:

<u>Foundation Widgets</u> :	<u>Layout Widgets</u> :	<u>Chart Widgets</u> :	<u>Other Widgets</u> :
• <u>Image</u> • <u>Multiple Images</u> • <u>Run Control</u> • <u>Run Status</u> • <u>OK/NG</u> • <u>Textbox</u> • <u>Button</u> • <u>Switch Control</u> • <u>Params Settings</u> • <u>Variable Settings</u> • <u>Character Settings</u> • <u>Traffic Light</u> • <u>Conditional Light</u>	• <u>Group Box</u> • <u>Child Interface</u> • <u>Tab Control</u>	• <u>Table</u> • <u>Line Diagram Array</u> • <u>Multi Line Diagram</u> • <u>Production Statistics</u> • <u>PieControl</u> • <u>Param Settings Array</u> • <u>Label Array</u>	• <u>Image</u> • <u>World Clock</u> • <u>Log</u>

Foundation Widgets

Image

You can subscribe and display the image output by a module in a flow through the image widget.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

Allows you to subscribe to the image source. You can click  and select an image output by a module.

Graphical Configuration

Allows you to select images for overlap display based on the data source. You can select images of multiple modules from the loaded flow.



Note

You need to set the **Data Source** first before setting this parameter. Otherwise, this parameter will be empty.

Overlay 1/2/3

Allows you to overlay other images for display based on images subscribed in the data source.



Note

- You need to first set the **Data Source** before setting this parameter.
 - You can select images output by a module only from the flow subscribed in **Data Source**.
-

Keep Last Result

Allows you to determine whether to display the last rendering result if no image is output in the subscribed result when the current flow ends.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Show StatusBar

Allows you to determine whether to display the status information (including image resolution, the current location of the cursor, and RGB or the gray value) in the lower right corner.

Show ToolBar

Allows you to determine whether to enable the tool bar (including functions of zooming in, zooming out, 1:1 ratio, full-screen, and locking) in the upper left corner.

Note

The 1:1 ratio can quickly restore to center the image display which is adaptive to the window size after the image is zoomed in, zoomed out, or moved.

Display Auxiliary Line

Allows you to determine whether to display the auxiliary cross line.

When there is no solution or procedure running, you can adjust the position of the auxiliary cross line by moving its center point.

List of historical images

When enabled, a list of historical images can be added below the image widget after you trigger the flow, The list displays the rendering results of the most recent flow executions.

Note

- The overlay layers are not displayed in the list of historical images.
 - When the image output rate is excessively high, a few images obtained from certain executions might be omitted.
-

Number of historical images

Allows you to set the maximum number of historical images to be displayed in the list.

Multiple Images

The multi-image widget has the same function as the image widget. The only difference is that this widget allows you to subscribe and display multiple images.

The specific parameters of this widget are divided into data source and layout.

- Data Source:

Image Quantity

Allows you to set the image quantity in the multi-image window and the location relationship among images.

Data Source

You can click **Config** to set the **PictureName**, **Image Source**, **Graphics Source**, **Overlay 1/2/3**, **Display Auxiliary Line** and **Keep Last Result**. The parameter settings are the same as that of the image widget.



Note

You are recommended to first set the **Image Quantity**.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click ☐ to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take

effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

Run Control

The running control widget can control the running method of all flows or a single flow. The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Sources:

Data Source

Allows you to subscribe to flows or AllProcess (namely the whole solution) in the loaded solutions.

Control Type

You can set the control type to **Once**, **Continuously**, or **Both Display**.

Once

Only the button for a single execution is displayed.

Continuously

Only the button for continuous execution is displayed.

Both Display

The above two buttons are displayed at the same time.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Button Color

Border Color

Icon Color

Allows you to set the color of the button, border, and icon.

Run Status

The running status widget can display the running status of all flows or a single flow. The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

Allows you to subscribe to flows or AllProcess (namely the whole solution) in the loaded solutions.

Note

- If you subscribe to all flows, when any flow in the solution is executed, the icon is displayed as "Running" (busy status). Otherwise, the icon is displayed as "Stop Run" (free status).
 - If you subscribe to a single flow, when the selected flow is executed, the icon is displayed as "Running" (busy status). Otherwise, the icon is displayed as "Stop Run" (free status).
-

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

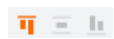


Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Free-Background Color

Free-Foreground Color

Busy-Background Color

Busy-Foreground Color

Allows you to set the color of each part for the icon during free status and busy status.

Fillet Radius

Allows you to set the fillet radius of four corners of the icon to control the icon shape.

OK/NG

The OK/NG widget can display OK (represented by 1) or NG (represented by 0) based on the subscribed data.

The specific parameters of this widget are divided into data source, layout, appearance, OK, and NG.

- Data Source:

Data Source

Allows you to subscribe to the int-type result of each module in a solution. You are advised to subscribe to the module status of each module and the result of the **Condition Detection** module.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

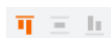


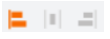
Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Fillet Radius

Allows you to set the fillet radius of four corners of the icon to control the icon shape.

- OK/NG:

OK-Content

NG-Content

Allows you to set the display content when the icon is in OK or NG status.

OK-Font

NG-Font

Allows you to set the font size and style when the icon is in OK or NG status.

OK-Border Thickness

NG-Font Thickness

Allows you to set the border thickness of the widget when the icon is in OK or NG status.

OK-Background

NG-Background Color

OK-Font Color

NG-Font Color

OK-Border Color

NG-Border Color

Allows you to set the color for the background, font, and border of the widget when the icon is in OK or NG status.

Textbox

The textbox widget can display the defined text information on the running window. You can customize the text content or subscribe to different types of data in a solution for display.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Type

Allows you to set the display data on the textbox to **string**, **int**, or **float** type.

Data Source

Allows you to subscribe to the module result, global variable, local variable, or solution path of each module in a solution.



Note

The data that can be subscribed to may vary according to the selected data type. You can subscribe to the solution path only when setting the **Data Type** to **string**.

Format String

This parameter is automatically loaded after you set the **Data Source** and is displayed as "Description for subscribed data: <subscribed data>" by default. You can customize the "Description for subscribed data:".

"<subscribed data>" is a formatted syntax, which varies in different data types.

- When you set **Data Type** to **string**, the formatted syntax is "%s".
- When you set **Data Type** to **int**, the formatted syntax is "%0d".

- › If the formatted syntax remains at its default content, the widget will output the int-type data of the data source.
- › If you have special needs, you can modify the Arabic numeral after "%" to set the maximum number of digits for the output int-type data. For example, if the original data is 600, and the formatted syntax is set to "%5d", then the widget will output 00600 when the solution is executed.
- When you set **Data Type** to **float**, the formatted syntax is "%1.3f". In this case, three effective numbers are remained behind the decimal point. You can customize the arabic numeral after "%". For example, "%1.6f" represents that six effective numbers are remained behind the decimal point.



Note

Ensure that you know the requirements related to formatted syntaxes when setting this parameter. If not, contact our technical support for more information.

Text

Displays information after you subscribe to a data source, and allows you to customize the display information if you do not subscribe to a data source.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click ☐ to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the

invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

 corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Font

Allows you to set the font size and style for the widget.

Font Color

Allows you to set the font color for the widget.

Button

You can use the button widget to set or redo some functions of the Software on the running window.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to open the configuration window, and then click  to add trigger commands.

You can set the **Type**, **Method**, and **Input Params** for each trigger command.



Note

You can configure multiple trigger commands for one button. When in use, the commands are executed for each click based on the set command sequence.

The trigger command types include interface link, parameter configuration, solution, execution, and reset.

Interface Link

Allows you to click the button to open a specific window. The supported windows include the **Global Camera** window, the **Global Communication** window, the **Global Variable** window, the **Log** window, and the **Comprehensive Configuration** window.

Param Config

Allows you to click the button to open the parameter configuration window of a flow or a module.

Solution

Allows you to click the button to save the current solution, or save it as a new solution.

Execute

Allows you to click the button to run flows once, start/stop continuous run, and execute module actions.

Run Once

Run all flows (**All Process**), a single flow, or a single module once.

Start Continuous Run

Stop Continuous Run

Start or stop the continuous run of all flows (**All Process**) or a single flow.

Module Action

Execute specific actions on a selected module. For example, you can reset variables in the **Variable Calculation** module or clear calibration points in the **N-point Calibration** module.

Reset

Allows you to click the button to reset a selected object. The objects that can be reset include **Variable Calculation** and **Data Queue**.

Variable Calculation

You are allowed to reset the variables of the variable calculation module in all flows (namely **AllCalculators**) or in a single flow.

Data Queue

You can reset all data queue modules (namely **AllDataQueueModules**).

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Show Text

Allows you to set the text to be displayed on the button,

Font

Allows you to set the font size and style for the widget.

Font Color

Background

Border Color

Click Color

Allows you to set the color of the font, background, border, and button.

ImageData

You can click  to select a image from your local path to overlay on this button.

Stretch Mode

Allows you to set the display mode of the imported **ImageData** on this widget.

Available options include None, Fill, Uniform, and UniformToFill.

Fillet Radius

Allows you to set the fillet radius of four corners of the icon to control the icon shape.

Switch Control

You can subscribe to the bool parameter through the switch control widget and determine whether to enable this parameter when running a solution on the running window.

The specific parameters are divided into data source and layout.

- Data Source:

Data Source

Allows you to subscribe to bool-type parameters in each module.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

 corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

Params Settings

You can subscribe to the int-type or float-type parameters of each module. After subscription, you can set the parameters when running a solution on the running window. The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Parameter Type

Allows you to set the type of the subscribed data to **int** or **float**.

Data Source

Allows you to subscribe to the int or float parameter in the solution.

Param

Allows you to customize the parameter name.

The Number Of Decimal Places

When the **Parameter Type** is set to **float**, you can adjust this value to determine the number of decimal places to be displayed.

- Layout:

Width

Allows you to set the width of this parameter widget. You can also adjust the widget width in the preview and edit area in real-time.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

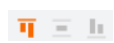


Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Background

Font Color

Allows you to set the color of the background and font for this widget.

Font

Allows you to set the font size and style for the widget.

Variable Settings

You can subscribe to the variables added in **Variable Calculation** modules. The variable will be marked as red when you subscribe to the expression. The variable can be displayed and modified on the running window in real time.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

Allows you to subscribe to variables added in the **Variable Calculation** module.

- Layout:

Width

Allows you to set the width of this parameter widget. You can also adjust the widget width in the preview and edit area in real-time.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Background

Font Color

Allows you to set the color of the background and font for this widget.

Font

Allows you to set the font size and style for the widget.

Character Settings

The character settings widget can display subscribed string information.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

Allows you to subscribe string-type variables of global/local variables and string-type parameters of each module.

- Layout:

Width

Allows you to set the width of this parameter widget. You can also adjust the widget width in the preview and edit area in real-time.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

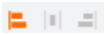
For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the

Align parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Background

Font Color

Allows you to set the color of the background and font for this widget.

Font

Allows you to set the font size and style for the widget.

Traffic Light

The traffic light widget can display the module status in real time.

You can determine the display color (red, represented by 0, or green, represented by 1) of the widget based on the subscribed module status.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

Allows you to subscribe to the status of each module.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Traffic Light Content

Allows you to customize the display content for the traffic light widget.

Font

Allows you to set the font size and style for the widget.

Font Color

Allows you to set the font color for the widget.

Conditional Light

You can determine the display color of the widget based on the subscribed data and the set conditions.

Note

If the subscribed parameter meets multiple conditions, the first color meeting the conditions is displayed based on the order of > value, < value, and default value.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Parameter Type

Allows you to set the type of the subscribed data to **int** or **float**.

Data Source

Allows you to subscribe to the int or float parameter in modules or global variables.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Traffic Light Content

Allows you to customize the display content for the conditional light widget.

Font

Allows you to set the font size and style for the widget.

Font Color

Allows you to set the font color for the widget.

Fillet Radius

Allows you to set the fillet radius of four corners of the icon to control the icon shape.

DefaultValueBackColor

DefaultValueBorderColor

>ValueBackColor

>ValueBorderColor

<ValueBackColor

<ValueBorderColor

Allows you to set the border/background color of the conditional light when the subscribed value is the default value, ">" value, or "<" value.

>ValueThreshold

<ValueThreshold

Allows you to set the threshold values for ">" and "<" conditions.



Note

You do not need to set the default value, which corresponds to the value that does not meet the ">" condition and "<" condition at the same time.

Layout Widgets

Group Box

You can group multiple widgets which were added to the Runtime Interface together through the group box widget.

You can move a widget into the group box by selecting and dragging the widget to the area where the group box is located.

Note

The method for removing widgets is the same as that for moving widgets into the group box.

The specific parameters of this widget are divided into layout and appearance.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

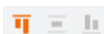
You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

 corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to customize the title displayed in the upper left corner of the group box widget.

Font

Allows you to set the font size and style for the widget.

Header Background

Font Color

Background

Border Color

Allows you to set the color for the header background, font, background, and border of this widget.

Child Interface

This widget allows you to add a new Runtime Interface as the sub-window to the current Runtime Interface. After configuration, a new Runtime Interface will pop up when you click this widget. The usage method for widgets in the subwindow is the same as that on the Runtime Interface.



Note

At most three levels of sub-windows can be embedded in the Runtime Interface.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Edit Subinterface

You can click **Config** to switch to the subinterface design window in the preview and edit area. In the right configuration pane of that window, you can adjust the layout, split template and basic settings of the widget. When you complete the configuration, you can close the subinterface design window and go back to settings of the Runtime Interface.

Password

You can click **Config** to determine whether to set a password for the child window. No password is set by default.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Text

Allows you to customize the name of the child window.

Font

Allows you to set the font size and style for the widget.

Font Color

Background

Border Color

Allows you to set the color for the font, background, and border of this widget.

Fillet Radius

Allows you to set the fillet radius of four corners of the icon to control the icon shape.

Display Icon

Allows you to determine whether to display  in front of the **Text** information in the child interface widget.

Tab Control

The tab control is the combination of multiple group box widgets. One group box corresponds to one tab.

Note

The method for moving widgets into or removing widgets from the **Tab Control** is the same as that of the ***Group Box***.

You can configure parameters for both the entire tab control widget and multiple contained tab pages.

- When you select the entire tab control widget, the specific parameters are divided into data source, layout, and appearance.

- Data Source:

Tab Page Quantity

Allows you to set the number of the contained tab pages.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

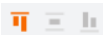
You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

 corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Background **Border Color** **Font Color**

Allows you to set the color for the background, border, and font of this widget.

Select Background

Allows you to set the color for the title bar of a tab page when the tab is selected.

Align

Allows you to set the alignment method for the title bar of tab pages. Available options include left alignment, top alignment, right alignment, and bottom alignment.

Display Icon

Allows you to determine whether to display the title bar of tab pages.

- When you select a single tab, the specific parameters are divided into layout and appearance.
 - Appearance:

Tab Page Name

Allows you to customize the name of the selected tab page.

- Layout:

Size

Allows you to view the tab size.



Note

You can modify the tab size in the preview and edit area.

Margin

Allows you to adjust the tab position. You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Chart Widgets

Table

This widget can present the subscribed data by scrolling through the columns, and historical results are presented in reversed order based on execution times. The table widget is usually used when you need to view the running results of multiple flows at the same time.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to set the column name, column width, data type, data source, and color for each column.



Note

You are recommended to set the number of **Columns** first.

Data Type

The data type can be set to string, int, or float.

Data Source

Allows you to subscribe to data meeting the requirement of the set **Data Type** in modules or global variables.

Color Edit

Allows you to set **Brush** for subscribed data in a table when the data meets the requirements of the set **Condition** and **Comparison Value**.



Note

If a piece of data meets multiple conditions, the color that corresponds to the first met condition takes effect.

Condition

The condition can be set to **=** (namely "equal to"), **!=** (namely "not equal to"), **>**, **<**, **>=**, and **<=**.



Note

String-type data only support **=** and **!=**.

Rows

Columns

Allows you to set the rows and columns of the table.

Line Title

Allows you to set the display content of the first cell in the upper left corner of the table.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Font

Allows you to set the font size and style for the widget.

Title Bar Content

Title Bar

Font Color

Background

Column header background

Row background

Alternating row background

Border Color

Allow you to set colors for different elements of the widgets, including the title and other fonts, backgrounds of title bar, column header and rows, as well as the border.

Line Diagram Array

This widget displays subscribed data with a linear relationship between the total execution times and the data corresponding to the current execution. Each data source is presented with a separate line graph.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to open the configuration window, and then click  to add data.

A piece of data consists of the following parameters.

Data Type

The data type can be set to int or float.

Data Source

Allows you to subscribe to the int or float parameter in the solution.

Name

Allows you to customize the data name.

Array Index

If you subscribe to an array as the **Data Source**, you can set this parameter to determine which element in the array to be displayed. The value is 0 by default, which means that the first element in the array is displayed. When the **Data Source** is not an array, this parameter should be set to 0.

Default Color

Allows you to set the color for each piece of data displayed in the widget.

UL Limits

You can click  to open the configuration window and then enable **Display Upper And Lower Limits** to set the parameters below.

Upper Limits

Lower Limits

Allows you to set the values and colors for the upper limit and lower limit.

The upper and lower limits will be displayed with the set color in the widget, which allows you to quickly judge whether the data is within the upper and lower limits.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

 corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Title Bar Content

Title Bar

Graph Background

Coordinate Axis Color

Allows you to set the color for each part of this widget.

Data of The Latest RunTimes

Allows you to control this widget to obtain the results of how many recent executions.

Multi Line Diagram

This widget displays subscribed data with a linear relationship between the total execution times and the data corresponding to the current execution. The difference from the **Line Diagram Array** is that the multi-line diagram widget allows to present multiple data sources in a single line graph.

For the parameter settings of this widget, refer to *[Line Diagram Array](#)*.

Production Statistics

This widget allows you to view the real-time value, maximum value, minimum value, average value, range and standard difference of the subscribed data.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to open the configuration window, and then click  to add data.

A piece of data consists of the following parameters.

Data Type

The data type can be set to int or float.

Data Source

Allows you to subscribe to the int or float parameter in the solution.

Name

Allows you to customize the data name.

Array Index

If you subscribe to an array as the **Data Source**, you can set this parameter to determine which element in the array to be displayed. The value is 0 by default, which means that the first element in the array is displayed. When the **Data Source** is not an array, this parameter should be set to 0.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

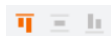


Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Show Columns Header

Allows you to select the displayed data. You can check the current value, maximum, minimum, average, range, and standard deviation.

Title Bar Content

Title Bar

Column header color

Column header background

Row background

Alternating row background

Font Color

Allows you to set the color for each part of this widget.

PieControl

This widget can sum the values of multiple data sources as a whole, and display the proportion of each value in the whole on the running window when you run the flow. The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to open the configuration window, and then click  to add data.

A piece of data consists of the following parameters.

Data Type

The data type can be set to int or float.

Data Source

Allows you to subscribe to the int or float parameter in the solution.

Name

Allows you to customize the data name.

Array Index

If you subscribe to an array as the **Data Source**, you can set this parameter to determine which element in the array to be displayed. The value is 0 by default, which means that the first element in the array is displayed. When the **Data Source** is not an array, this parameter should be set to 0.

Default Color

Allows you to set the color for each piece of data displayed in the widget.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click ☐ to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Title Bar Content

Title Bar

Background

Allows you to set the color for each part of this widget.

Param Settings Array

This widget is a combination of multiple **Params Settings** widgets.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to open the configuration window, and then click  to add data.

A piece of data consists of the following parameters.

Param

Allows you to customize the parameter name.

Data Type

Allows you to set the type of the subscribed data to **int** or **float**.

Data Source

Allows you to subscribe to the int or float parameter in the solution.

Decimal Digits

When the **Data Type** is set to **float**, you can adjust this value to determine the number of decimal places to be displayed.

Orientation

The added parameters can be arranged horizontally or vertically.

Spacing

Allows you to set the spacing between multiple widgets in the **Param Settings Array** widget.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

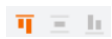


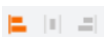
Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- **Appearance:**

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Font

Allows you to set the font size and style for the widget.

Title Bar Content

Title Bar

Font Color

Allows you to set the color for each part of this widget.

Label Array

This widget is a combination of multiple **Textbox** widgets.

The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

Data Source

You can click **Config** to open the configuration window, and then click  to add data.

A piece of data consists of the following parameters.

Text

Displays information after you subscribe to a data source, and allows you to customize the display information if you do not subscribe to a data source.

Format String

This parameter is automatically loaded after you set the **Data Source** and is displayed as "Description for subscribed data: <subscribed data>" by default. You can customize the "Description for subscribed data:".

"<subscribed data>" is a formatted syntax, which varies in different data types.

- When you set **Data Type** to **string**, the formatted syntax is "%s".
- When you set **Data Type** to **int**, the formatted syntax is "%0d".
 - › If the formatted syntax remains at its default content, the widget will output the int-type data of the data source.
 - › If you have special needs, you can modify the Arabic numeral after "%" to set the maximum number of digits for the output int-type data. For example, if the original data is 600, and the formatted syntax is set to "%5d", then the widget will output 00600 when the solution is executed.
- When you set **Data Type** to **float**, the formatted syntax is "%1.3f". In this case, three effective numbers are remained behind the decimal point. You can customize the arabic numeral after "%". For example, "%1.6f" represents that six effective numbers are remained behind the decimal point.



Note

Ensure that you know the requirements related to formatted syntaxes when setting this parameter. If not, contact our technical support for more information.

Data Type

Allows you to set the display data on the textbox to **string**, **int**, or **float** type.

Data Source

Allows you to subscribe to the module result, global variable, local variable, or solution path of each module in a solution.

Note

The data that can be subscribed to may vary according to the selected data type. You can subscribe to the solution path only when setting the **Data Type** to **string**.

Color Edit

Allows you to set **Brush** for subscribed data in a table when the data meets the requirements of the set **Condition** and **Comparison Value**.

Note

If a piece of data meets multiple conditions, the color that corresponds to the first met condition takes effect.

Condition

The condition can be set to **=** (namely "equal to"), **!=** (namely "not equal to"), **>**, **<**, **>=**, and **<=**.

Note

String-type data only support **=** and **!=**.

Orientation

The added parameters can be arranged horizontally or vertically.

Spacing

Allows you to set the spacing between multiple widgets in the **Label Array** widget.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align

   corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And    correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Font

Allows you to set the font size and style for the widget.

Title Bar Content

Title Bar

Font Color

Allows you to set the color for each part of this widget.

Other Widgets

Image

This widget allows you to import local pictures and display them on the running window. The specific parameters of this widget are divided into data source, layout, and appearance.

- Data Source:

ImageData

You can import pictures from the local path.

Stretch Mode

Allows you to set the display mode of the imported **ImageData** on this widget. Available options include None, Fill, Uniform, and UniformToFill.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Background

Border Color

Allows you to set the color for the picture background and border.

Border Thickness

Allows you to set the border thickness for the picture.



Note

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

World Clock

This widget can display the current time of the selected time zone.

The specific parameters of this widget are divided into time setting, layout, and appearance.

- Time Setting:

Format

Allows you to set the format type for displaying the time.

Time Zone

Allows you to set the time zone.

- Layout:

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.



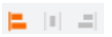
Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Font

Allows you to set the font size and style for the widget.

Font Color

Allows you to set the font color for the widget.

Log

This widget can display the log information of the Software on the running window in real time (namely the **Log** of the **System**).

The specific parameters of this widget are divided into layout and appearance.

- Layout:

Auto Fill

Allows this widget to fill the entire Runtime Interface automatically.

Size

Allows you to adjust the widget size. You can also adjust the widget size in the preview and edit area.

Margin

Allows you to adjust the widget location. You can also adjust the location in real-time in the preview and edit area. The margin consists of the values of four parameters T, B, L, and R, which correspond to the pixel numbers of the distance from the widget to the top, bottom, left side, and right side of the Runtime Interface.

You can click  to the right of either the T, B, L, or R parameter to update all values to the value of the selected parameter.

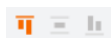


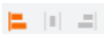
Note

For this parameter, only one of T and B and one of R and L take effect, which depends on the selected vertical alignment method and horizontal alignment method in the **Align** parameter.

For instance, if you select **Top Align** and **Left Align**, the point in the upper left corner is selected as the origin for the **Margin** parameter. In this case, the parameters that take effect are T and L, which correspond to the distance from the top and the left side of this widget to the top and the left side of the Runtime Interface. The values of the invalid parameters B and R are displayed as 0, and the manual modification does not work.

Align



corresponds to top alignment, center alignment, and bottom alignment in the vertical direction, respectively. And  correspond to left alignment, center alignment, and right alignment in the horizontal direction, respectively.

- Appearance:

Title

Allows you to set the title name.

Display TitleBar

Allows you to determine whether to display the title bar.

Title Bar Content

Title Bar

Column header color

Column header background

Row background

Alternating row background

Font Color

Allows you to set the color for each part of this widget.